



Minimal filesystem === Basic applications

- ▶ In order to work, a Linux system needs at least a few applications



An `init` application, which is the first user space application started by the kernel after mounting the root filesystem (see <https://en.wikipedia.org/wiki/Init>):

- The kernel tries to run the command specified by the `init=` command line parameter if available.
- Otherwise, it tries to run `/sbin/init`, `/etc/init`, `/bin/init` and `/bin/sh`.
- In the case of an `initramfs`, it will only look for `/init`. Another path can be supplied by the `rdinit=` kernel argument.
- If none of this works, the kernel panics and the boot process is stopped.



- The `init` application is responsible for starting all other user space applications and services, and for acting as a universal parent for processes whose parent terminate before they do.
- ▶ A shell, to implement scripts, automate tasks, and allow a user to interact with the system
- ▶ Basic UNIX executables, for use in system scripts or in interactive shells: `mv`, `cp`, `mkdir`, `cat`, `modprobe`, `mount`, `ip`, etc.
- ▶ These basic components have to be integrated into the root filesystem to make it usable



Overall booting process

